

Name: _____

In the first diagram, $\overline{PR} \perp \overline{QR}$, $PQ = 2x - 1$, and $PR = 17$.

1. Name the shortest segment from point P to line QR.
2. Write and solve an inequality for x.

In each diagram, the two marked angles form a right angle. Find x.

3. (figure a)
4. (figure b)

Answer each question and name the theorem that supports your answer.

5. In a plane, line $a \perp$ line c and line $b \perp$ line c . What can you conclude about a and b ?
6. In a plane, line $r \parallel$ line s and line $t \perp$ line r . What can you conclude about t and s ?
7. Two lines intersect to form a linear pair of congruent angles. Find the measure of each angle and describe the lines.

Use coordinates.

8. Find the distance from point $A(-2, 4)$ to the line $y = 1$.
9. Segment CD has endpoints $C(1, 3)$ and $D(9, 3)$. Write an equation of the perpendicular bisector of $\overline{CD}$.

Complete the two-column proof. (Use the last diagram, where $j \parallel k$.)

10. Given: $j \parallel k$, $\angle 1 \cong \angle 2$ Prove: $t \perp k$

Statements	Reasons
1. $\angle 1 \cong \angle 2$	1. Given
2.	2.
3.	3.
4.	4.
5.	5.

